

Classification of Leukemia and Lymphoma using Faster R-CNN

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Abstract— *Blood disorders are disorders that impair the ability of the blood to function normally. There are various types such as leukemia and lymphoma, and the symptoms differ accordingly. Many death cases are related to these malignant tumors as a result of the difficult therapy and its negative effects. Furthermore, patients require a variety of diagnostic tests, which are costly, time-consuming, and can be stressful and hazardous to the patients. Furthermore, only a qualified pathologist can distinguish and identify various forms of lymphomas and leukemias. In this paper, the “Faster Region-Based Convolutional Neural Network; Faster R-CNN” algorithm is implemented to classify normal patients and ill patients into four categories: Acute Lymphoblastic Leukemia (ALL), Chronic Lymphocytic Leukemia (CLL), Follicular Lymphoma (FL), and Mantle Cell Lymphoma (MCL). This is done by dealing with Region of Interest (ROI) convolutional layer, then applying Vgg-16 model & Region Proposal Network (RPN) layer and adding a classifier layer. The dataset is based on microscopic and biopsies images. The proposed model achieved an accuracy of 87,575% after training in an acceptable computational time.*

Keywords— *Blood disorders; leukemia; lymphoma; Faster R-CNN; Acute Lymphoblastic Leukemia; Chronic Lymphocytic Leukemia; Follicular Lymphoma; Mantle Cell Lymphoma; normal patients.*

I. INTRODUCTION

Blood is a watery tissue that connects and flows around the heart and blood arteries through the lymphatic system of the body. All blood cells start out as pluripotent stem cells and go through multiple stages of development before becoming unique cells of various sorts. Plasma, red blood cells, white blood cells, and platelets are the four major components of the blood [1].

Blood diseases are abnormalities that impact the capacity of the blood to operate properly. Blood disorders are divided into two main categories; the first one, causes a reduction in blood and its components and hinders their function, such as anemia if the condition includes erythrocytes; leukopenia if the disorder involves leukocytes, and thrombocytopenia if the disorder involves platelets. And the second one, causes a rise in blood and its components such as erythrocytosis, which affects red cells, leukocytosis, which affects white blood cells, and thrombocythemia or thrombocytosis, which affects platelets [2].

Red blood cells, or erythrocytes, are the functional part of the circulatory system that carry gases and nutrients all through the body [3]. Platelets are fragments of

megakaryocytes, which are very large cells in the bone marrow. They promote in the production of red blood cells clots, which slow or halt bleeding and aid wound healing [4]. Leukocytes, commonly known as White Blood Cells, aid in the body's defense against foreign substances, bacteria, and infectious illnesses. The composition and concentration of white blood cells in the blood provide significant information and play an important role in the diagnosis of many disorders. As a result, white blood cell detection is the very first module in the proposed system [5]. The main diseases related to WBCs and which lead to cancer and tumorous blood conditions, in this research, are leukemia and lymphoma. Both cancers develop from the same type of abnormal WBC but their origins, causes, numerous subtypes, symptoms, and treatments make them separate conditions.

1- Leukemia

Leukemia is one of the lists of top 10 most dangerous cancers that jeopardize human life. Leukemia affects the oxygen carrying capacity of blood as well as its other functions. It is a bone marrow condition that develops when an aberrant white blood cell continues to duplicate itself indefinitely. Leukemic cells move through the circulation and continue to divide, occasionally creating tumors and causing damage to organs such as the kidney and liver [6].

The common types of leukemia are: Chronic lymphocytic leukemia (CLL), Acute myeloid leukemia (AML), Chronic myeloid leukemia (CML) and Acute lymphocytic leukemia (ALL). Despite improvements in oncology, survival rates following treatment of leukemia remain remarkably low due to severe life-threatening health consequences of chemotherapeutic drugs and tumor reappearance. Specific treatment drugs are used to treat different forms of leukemia. Each patient will receive a customized treatment regimen. Age, overall health, kind of leukemia and other distinctive aspects of the disease, response to first therapy, and a variety of other factors all influence treatment. The healthcare team will devise a treatment plan that they think will be the most effective [7].

2- Lymphoma

Lymphoma is a kind of cancer that starts in the immune system's infection-fighting cells, known as lymphocytes. When you have lymphoma, your lymphocytes alter and expand uncontrollably. Every variety of lymphoma develops at a different rate and reacts to therapy differently. The types of lymphoma, that affect lymph nodes; the Follicular Lymphoma (FL) and the Mantle Cell Lymphoma (MCL) [8].

3- Diagnosis of Leukemia and Lymphoma

Both leukemia and lymphoma are diagnosed with the same methods, the physician will ask for physical exam, order

complete blood count (CBC) test. If these diagnostic results are suspicious, imaging tests and bone marrow biopsies are essentials.

The methods are explained as follows [8]:

- Physical exam: Patient's symptoms and enlarged lymph nodes are inspected. The gums, bruises or red skin rash, as well as enlarged spleen are examined.
- Complete blood count (CBC): In case of leukemia, red blood cell and platelet counts will be lower than normal, but white blood cell count will be greater than usual. Other tests may be used to detect chromosomal abnormalities and other markers on cells that aid in determining the kind of leukemia or lymphoma.
- Imaging and other tests: A diagnosis can be done using chest X-ray, CT scan, or magnetic resonance imaging (MRI) scan. Moreover, a lumbar puncture may be performed to determine whether the cancer has progressed to the spinal fluid surrounding the brain and spinal cord.
- Bone Marrow Biopsy (Marrow Aspiration): a large needle is used to extract fluid from the bone marrow, generally from a location around the hip (pelvic bone). The blood cells in the fluid are then examined under a microscope. The measured proportion of aberrant cells in the bone marrow confirm the presence of leukemia or lymphoma diagnosis.

II. RELATED WORKS

Leukemia is a cancer that begins in blood-forming tissue, such as bone marrow, and causes an irregularly large number of blood cells to be produced and enter the circulatory system. In conventional methods, hematologists use blood and bone marrow tests to diagnose ALL. Manual blood testing techniques that have been used for a long time are often slow and produce less accurate results.

Many previous studies based on machine learning are done to detect these types of diseases. One of the important studies helps in ALL diagnosis using a computer-aided system that produces accurate results by utilizing image processing and deep learning techniques. Researchers proposed a method for classifying ALL into subtypes and reactive bone marrow in stained bone marrow images. To achieve accurate classification results, a robust segmentation and deep learning techniques with the convolutional neural network CNN are used to train the model on bone marrow images. They only consider the slides of patients with a history of ALL, and some images of normal or reactive bone marrow from patients with no history of leukemia are also taken; the slides are stained with Leishman stain and examined under a microscope. With the assistance of hematologists, the images are taken and classified into three subtypes of ALL: L1 (100 images), L2 (100 images), L3 (30 images), and normal (100 images), resulting in the formation of a dataset [10].

Different feature and classifier combinations are trained and tested for classification. This method is a deep learning technique based on the AlexNet model and CNN for classifying ALL into subtypes and normal conditions. The

model is configured based on the data, and the last three layers are fully connected, softmax, and classification layer are fine-tuned to the new classifier layers and are set based on the new data. For performance evaluation, the data sets are also tested on other classifiers, such as k-Nearest Neighbors (KNN), naive Bayesian, and Support Vector Machine; (SVM). The technique proved to be 97.78% accurate. And the other evaluations give; Bayesian accuracy is 78.34%, KNN and SVM are respectively 80.42% and 90.91%.

Hematological diseases, such as leukemia and lymphoma, cause many deaths due to the hard treatment and its side effects, also to the late detection of those illnesses is another reason for death. Moreover, patients need many types of diagnostic tests to confirm the cancer presence as blood test, bone marrow biopsy, urine tests, spinal tap. Whereas, in most cases, patients need to perform imaging tests such as X-ray, CT, or PET scans, MRI, and ultrasound to confirm the presence of cancer. Those tests also are costive, time intensive and can be very exhausting and harmful to the patients. Furthermore, only specialized pathologist can discriminate between these types of lymphomas and leukemias and to classify them. Moreover, the similar studies detected or compared between two types of these WBC diseases.

Thus, this paper will show a design of an algorithm that will be efficient to the diagnosis and classification of the WBC blood diseases; Follicular Lymphoma (FL), Mantle Cell Lymphoma (MCL), Acute Lymphoblastic Leukemia (ALL) and Chronic Lymphocytic Leukemia (CLL) as shown in figure 1. While this study is helpful with time and cost reduction, it will also be supportive to the pathologist while taking the decision about the tumor and its treatment.

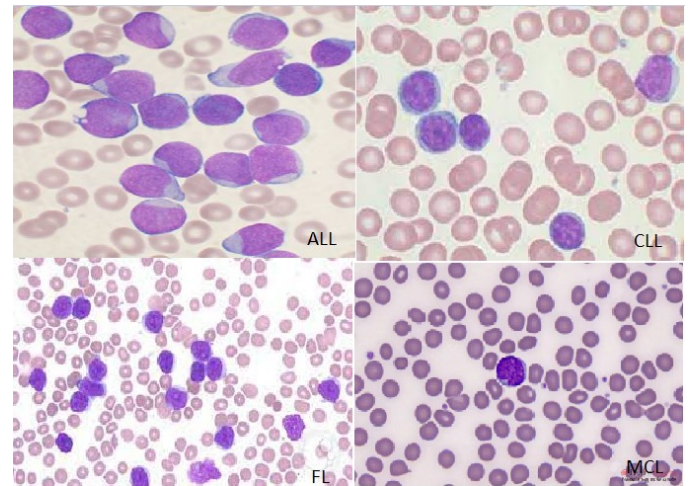


Fig. 1. Morphology of different WBC illnesses

This paper is organized as follows. Section II presents related work. Section III summarizes the dataset and used network model. Section IV provides the results and a discussion on the contribution. The conclusion is provided in section V and future work in section VI.

III. MATERIALS & METHODS

Machine learning is a field of research which focuses on the use of data and algorithms to imitate the way that humans learn. It is a type of artificial intelligence AI that allows software applications to become more accurate at predicting

outcomes. Thus, to apply our model we are in need of a dataset.

The team here has suggested to take 15,135 microscopic images for leukemia and 374 biopsies images for lymphoma (139 for FL, 113 for CLL and 122 for MCL) as dataset. The dataset is retrieved from Kaggle platform and it is from 118 patients with two named classes: Leukemia blast and normal cell. The ALL is classified by identifying immature leukemic blasts from normal cells from this dataset. This dataset is a collection of samples prepared by different pathologists at different sites. A data augmentation will be applied to the small dataset to have a balanced dataset of the diseases.

This dataset will be an input for a network model which is the faster R-CNN to extract features from the image dataset and classify the blood disease into ALL, CLL, FL, and MCL as shown in the below chart (see figure 2).

One of the most well-known object identification neural networks is Faster-RCNN. It also serves as the foundation for numerous derived networks for segmentation, detection, feature extraction, and so forth. Faster-RCNN is fundamentally consists of three neural networks: Feature Network, Region Proposal Network (RPN), and Detection Network as shown in figure 3. The Feature Network is a well-known pre-trained image categorization network. This network's function is to extract useful characteristics from images. This network's output retains the original image's form and structure. The RPN is thus typically a basic network with three convolutional layers. One common layer feeds into two layers: one for classification and one for bounding box regression. RPN's goal is to construct a set of bounding boxes known as Regions of Interest (ROIs) that have a high likelihood of including any item. Finally, the Detection Network combines both the Feature Network and the RPN as input and creates the final class and bounding box. It is often made up of four Fully Connected or Dense layers. A classification layer and a bounding box regression layer share two stacked common layers. The features are trimmed according to the bounding boxes to assist it categorize just the inside of the bounding boxes.

Google Colaboratory is a cloud platform that is used to implement this study. The model is set in an environment of several libraries, such as "`__future__` module" that is used to inherit new features that will be available in future Python versions. "Random" defines a series of functions for generating or manipulating random integers; "pandas" is used for data processing, inspection, and modification, and it includes charts and series, "keras" is used to make the implementation of neural networks easy. "Tensorflow" is a Python library for fast numerical computing; "Pprint" is a Python tool for printing data structures in a visually appealing manner. "Sys" provides access to system-specific parameters and functionalities, while "numpy" contains several helpful functions for doing mathematical or scientific computations. "Matplotlib" is used to create static, animated, and interactive infographics. "Optparse" modifies characteristics of the options object produced by parsing `args()` depending on command-line values as it parses the command line. "Pickle" converts any type of Python object (list, dict, etc.) into byte streams (0s and 1s), "CV2" is used as a tool in image processing and to perform computer vision tasks, "math" gives

access to several common math functions and constants that may be used throughout the code to do more advanced mathematical tasks, etc...

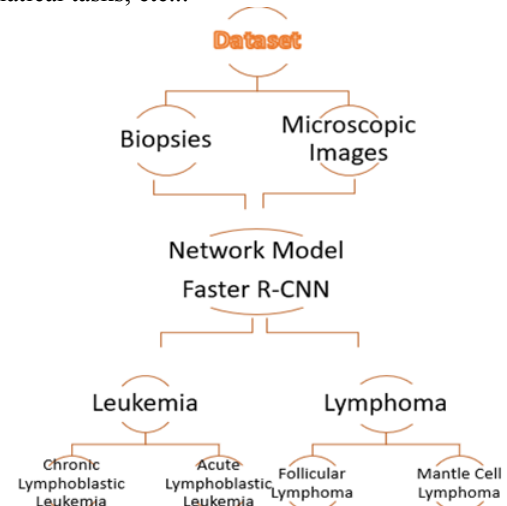


Fig. 2. Flowchart of proposed system.

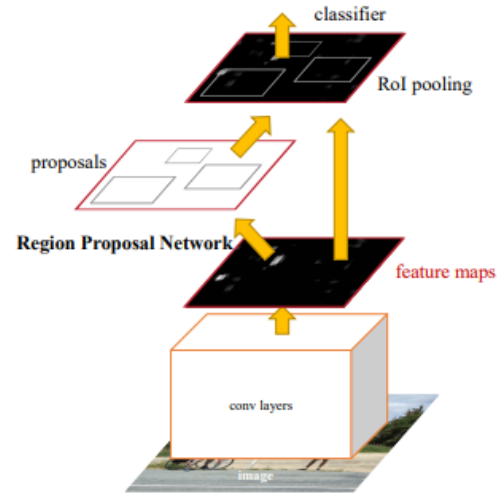


Fig. 3. Faster R-CNN architecture.

The performed steps are as follows:

- Mounting Drive to Google Colab and Importing Libraries.
- Setting the configuration and initialization of the classes, the global variables for data augmentation and variables related to the base network (vgg), anchor boxes, sample size, ROI and RPN settings and so on.
- The "Data Augmentation" consists of 'horizontal flips'; 'vertical flips' and '90 degree rotation'.
- Parsing and mapping the dataset in order to turn it into readable arrays.
- Pattern detection using ROI pooling convolutional layer. This layer is responsible of drawing the markers on all the potential regions of interest, as the name states. This will also help guide the algorithm with the identification of targeted areas in the dataset.
- Implementing the VGG-16 model which quantifies the results acquired from the previous and runs it through this pretrained algorithm, which will help translate the output of the ROI pooling into data usable by the subsequent RPN layer generator. This is done through the creation of five blocks using a convolution and MaxPooling which is

a pooling function that computes the maximum value for a feature map's patches and utilizes it to produce a down-sampled (pooled) feature map.

- Then the RPN layer code section is used to define and implement the boxing around the targeted areas, which gives a clearer view of the affected area in the form of squares drawn on the initial input picture.
- The last layer of this detection algorithm is the classifier layer, which is used to finalize the parsing and detection of patterns using faster R-CNN. It will be responsible of convolving the ROIs, flattening the convolution layer, and then returns the output class which is a softmax activation function to classify the class name of the object and the output regression which is responsible of the linear activation function for bounding boxes (bboxes) coordinates regression.
- After the parsing and detection process has finished, the Intersection of Union (IoU) calculation is done alongside the RPN anchor calculation of the input, to finalize and check the detected areas in the samples proposed. The IoU is used to find the intersecting regions and calculate their area.
- The RPN calculations are done to generate bboxes that indicate whether the RPN anchor should be a target. The ground-truth layer is thus generated. Based on these calculations, it will then be used to verify the integrity and "truthiness" of the output results, and thus ultimately increase the accuracy of the output.
- After that the losses functions are applied in order to compare the actual results in regards to the predicted ones.
- Finally, the model is built based on the base network (vgg-16) and the RPN layer (built on the base layers) which enables the program to save the weights and records, and output accurate plots of said result.
- All these steps are executed to generate the output with great accuracy, and display the results in terms of regression plots and boxes surrounding the regions of interest.

IV. RESULTS & DISCUSSION

Regarding the results of the Faster R-CNN algorithm; the accuracy escalated from epoch to another will be shown first. Then how the losses decreased from one layer to another and the mean overlapping bounding boxes will be presented.

In this study, before the training, we have the parsing step where the dataset is loaded in the training, which has taken around 105.58 mins, and the positives anchors (the purple and white lines) and the ground-truth bounding boxes (the green ones) are set as shown in figure 4 below. The anchor boxes are preset boundary boxes with a specific height and width. These boxes are often chosen based on the item sizes in the training datasets to capture the scale and aspect ratio of various object classes that the users wish to identify. However, Ground-truth bounding boxes are rectangular areas that surround the targeted pattern and were designated for training [11]. Figure 4 shows the targeted areas needed for the classification.

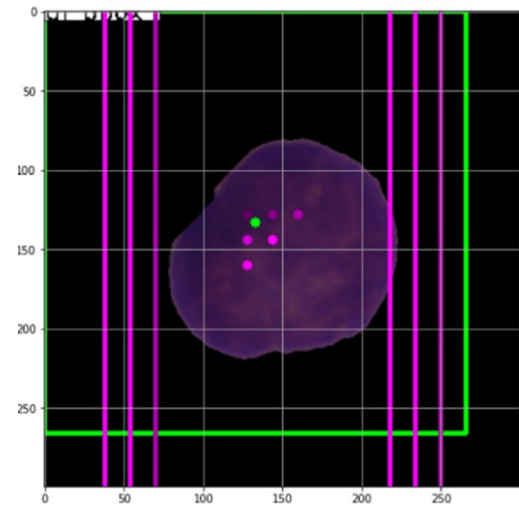


Fig. 4. Ground-truth bboxes and anchors.

Then, the training was done by 30 epochs, each epoch has 1000 as a length. Each epoch took around 4 minutes to be trained on the Colab GPU. The whole training took around two hours. For each epoch there are some calculations done; mean number of bounding boxes from RPN overlapping ground-truth boxes, the elapsed time, classifier's accuracy for each bounding box, loss RPN regression, loss RPN classifier, loss detector classifier, loss detector regression and the total loss and how much the loss decreased from epoch to another. From epoch 1 to 30, we start with an accuracy of 71.35% and with a total loss of 0.925814 to attain at the 30th epoch with an increased accuracy of 87.575% and the total loss decreased to 0.426399 as shown Figure 5.

```
Epoch 30/30
1000/1000 [=====] - 255s 255ms/step - rpn_cls: 2.3668e-05 -
Mean number of bounding boxes from RPN overlapping ground truth boxes: 8.103
Classifier accuracy for bounding boxes from RPN: 0.87575
Loss RPN classifier: 2.603751952614759e-05
Loss RPN regression: 0.018174507138319314
Loss Detector classifier: 0.3093854225985706
Loss Detector regression: 0.09881350975111126
Total loss: 0.42639947700752734
Elapsed time: 254.84448528289795
Total loss decreased from 0.4310494125684716 to 0.42639947700752734, saving weights
```

Fig. 5. Last epoch training summary.

At the end of the training, the indexes of the accuracy, those losses, and the mean overlapping bounding boxes are saved in an excel records, and a generation of graphs is done through the statement "plot" in the code.

Figure 6 shows the generated mean overlapping bboxes and the classifier's accuracy. The mean overlapping bboxes is when the predicted bounding box overlaps with the ground-truth bounding box. When the overlapping is more than 50% it is a reliable result. In addition, the accuracy of our model is the percentage of correct predictions that the classifier has reached.

The first term is the classification loss across two classes, whether or not there is an object. The second term, though, is the regression loss only when there is an object. The generation of those losses in the RPN layer are shown in Figure 7 and the losses generated at the level of the classifier are shown in Figure 8. These losses represented the difference between the true value and predicted value.

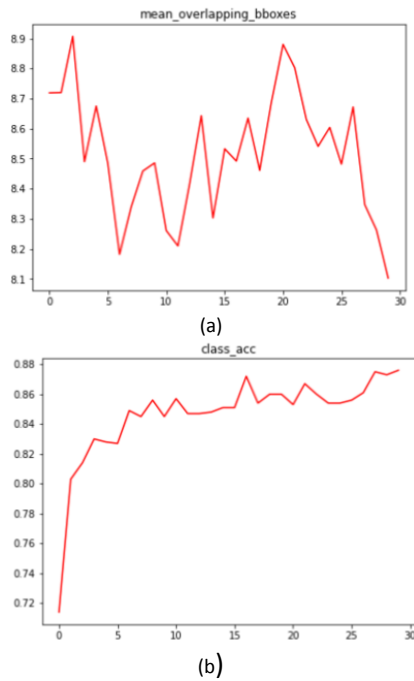


Fig. 6. Mean overlapping bboxes (a) and the accuracy(b).

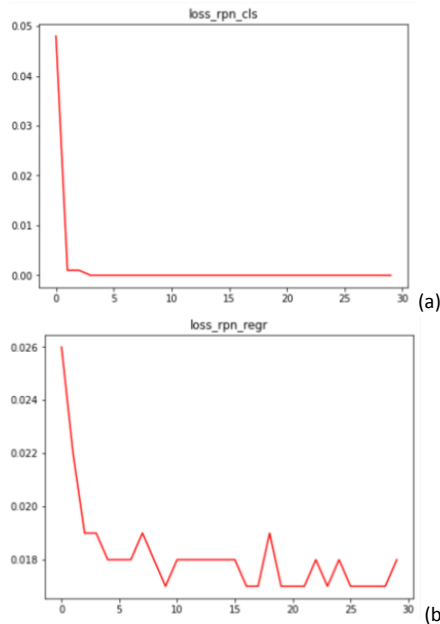


Fig. 7. Loss of the RPN classifier (a) and loss of the RPN regression (b).

The training was successful and attained the aim of this thesis with an accuracy of 87.575%, with generation of different plots which are shown above. The dataset augmentation helped us to have a better accuracy while our dataset was not that large.

V. CONCLUSION & FUTURE WORK

By comparing several research that utilized “CNN and SVM to identify just one or a maximum of two types of blood illnesses, we proposed working on a more sophisticated classification with a new deep learning approach. In this project, we attempted to build the “Faster R-CNN” algorithm, a deep learning approach capable of classifying several forms of lymphoma and leukemia as well as a healthy patient. Faster R-CNN is very efficient for real-time detection, very quick

computational time with respect to the CNN. After all our work, we attained an accuracy of 87,575% which is approximately 88%.

To increase the accuracy of our work in the future, we will require a diverse mix of datasets. Furthermore, the model may be trained to recognize various blood illnesses and categories of blood diseases; white blood cells, red blood cells and platelets diseases.

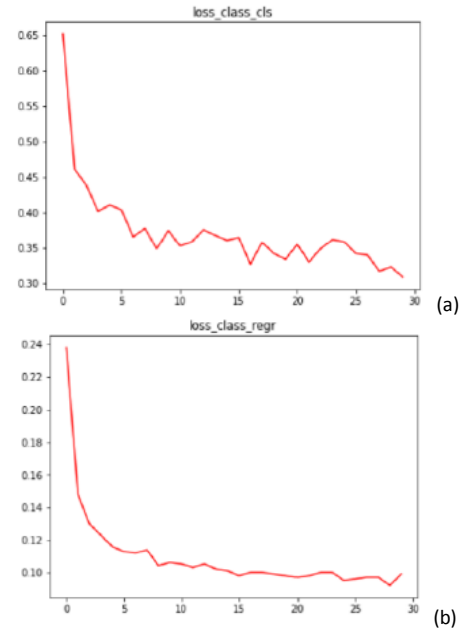


Fig.8. Loss of detector classifier (a) and loss of detector regression (b).

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